

Enhanced Private Pilot Certification Course**LESSON 145b (1.5 DUAL and 0.3 INSTRUMENT)****Lesson Objective**

During this lesson, the student will be asked to demonstrate private pilot proficiency in all areas of operation.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Dual)**Go/No-go Decision**

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Preflight Procedures

- Preflight inspection:
 - Ensure that the student conducted a thorough preflight inspection.
 - Walk around the airplane and quiz the student on:
 - Antennas.
 - Aircraft structure.
 - Flight controls.
 - Systems.
 - Placards and markings.
- Cockpit management:
 - Ensure the student came prepared with all of the necessary documents and equipment (kneeboard, headset, airport diagram, checklists, etc.)
 - Verify that the student can stay organized throughout the flight.
- Engine starting:

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- Verify correct procedures.
- Taxiing:
 - Verify correct procedures.
 - Ensure that aileron correction is established correctly.
 - No riding the brakes!
- Runway incursion avoidance:
 - Emphasize that it is the pilot in command's responsibility to avoid traffic.
 - IF YOU SEE SOMETHING, SAY SOMETHING!
- Before takeoff check:
 - Ensure that the student is using the correct checklist and verbalizing all items.
 - No items should be skipped.
 - Ensure that the student is performing the correct action for "Read and Do" vs "Do and Verify" checklists.

Airport Operations

- Radio communication and ATC light gun signals:
 - Radio communications must be made by the student.
- Traffic patterns:
 - Verify correct airspeeds.
 - Verify correct altitude.
 - Ensure that the student corrects for the wind.
- Airport, runway, taxiing signs, markings, and Lighting:
 - Quiz the student on various signs, markings, etc. during taxi.

Takeoffs, Landings, and Go-around

- Normal and crosswind takeoff and climb
 - Student says, student does.
 - Verify correct procedure.
 - Ensure proper crosswind correction.
 - Standards:
 - Rotate and lift off at the recommended airspeed and accelerate to V_Y .
 - Establish a pitch attitude to maintain the manufacturer's recommended speed or $V_Y + 10/-5$ knots.
 - Maintain $V_Y + 10/-5$ knots to a safe maneuvering altitude.
 - Maintain directional control and proper wind-drift correction throughout takeoff and climb.
- Normal and crosswind approach and landing
 - Student says, student does.
 - Verify correct procedure.
 - Ensure proper crosswind correction.
 - Standards:

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- Select and aim for a suitable touchdown point considering the wind, landing surface, and obstructions.
 - Establish the recommended approach and landing configuration and airspeed and adjust pitch attitude and power as required to maintain a stabilized approach.
 - Maintain directional control and appropriate crosswind correction throughout the approach and landing.
 - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.
- Short-field takeoff and maximum performance climb
 - Student does, student says:
 - Verify correct procedure.
 - Ensure that the student holds the brakes until full static RPM.
 - Verify that the climb out is conducted at 63 knots.
 - Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.
 - Standards:
 - Rotate and lift off at the recommended airspeed and accelerate to the recommended obstacle clearance airspeed or VX, +10/-5 knots.
 - Establish a pitch attitude that will maintain the recommended obstacle clearance airspeed or VX, +10/-5 knots until clearing the obstacle or until the airplane is 50 feet above the surface.
- Short-field approach and landing
 - Student does, student says:
 - Verify correct procedure.
 - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
 - Verify simulated maximum braking.
 - Standards:
 - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
 - Touch down at a proper pitch attitude within 200 feet beyond or on the specified point.
- Soft-field takeoff and climb
 - Student does, student says:
 - Verify correct procedure.
 - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
 - Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.

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- Verify that the student says positive rate of climb before retracting the flaps.
 - Standards:
 - Establish and maintain a pitch attitude that will transfer the weight of the airplane from the wheels to the wings as rapidly as possible.
 - Lift off at the lowest possible airspeed and remain in ground effect while accelerating to VX or VY, as appropriate.
 - Establish a pitch attitude for VX or VY, as appropriate, and maintain selected airspeed +10/-5 knots during the climb.
- Soft-field approach and landing
 - Student does, student says:
 - Verify correct procedure.
 - Verify minimal braking.
 - Standards:
 - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
 - Make smooth, timely, and correct control inputs during the round out and touchdown, and, for tricycle gear airplanes, keep the nose wheel off the surface until loss of elevator effectiveness.
 - Touch down at a proper pitch attitude with minimum sink rate, no side drift, and with the airplane's longitudinal axis aligned with the center of the runway.
 - Maintain elevator as recommended by manufacturer during rollout and exit the "soft" area at a speed that would preclude sinking into the surface.
- Forward slip to a landing
 - Student says, student does.
 - Verify correct procedure.
 - Power must be at idle.
 - Ailerons into the wind.
 - Rudder opposite.
 - Standards:
 - Configure the airplane correctly.
 - As necessary, correlate crosswind with direction of forward slip and transition to sideslip before touchdown.
 - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.
 - Maintain a ground track aligned with the runway center/landing path.
- Go-around/rejected landing
 - If the student has an unstable approach, see if they decide to initiate a go-around.
 - If all of the approaches are stable, have the student perform a go-around due to "something" on the runway (airplane, animal, vehicle, etc.)

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- Verify correct procedures.
- Standards:
 - Apply takeoff power immediately and transition to climb pitch attitude for VX or VY as appropriate +10/-5 knots.
 - Configure the airplane after a positive rate of climb has been verified or in accordance with airplane manufacturer's instructions.
 - Maintain VY +10/-5 knots to a safe maneuvering altitude.
 - Maintain directional control and proper wind-drift correction throughout the climb.

Performance Maneuver

- Steep turns:
 - Student does, student says:
 - Verify correct procedure.
 - Ensure that the student is looking both outside and inside.
 - Verify the roll out happens at ½ the bank angle before the desired heading.
 - Standards:
 - Roll into a coordinated 360° steep turn with approximately a 45° bank.
 - Perform the Task in the opposite direction, as specified by evaluator.
 - Maintain the entry altitude ±100 feet, airspeed ±10 knots, bank ±5°, and roll out on the entry heading ±10°.

Ground Reference Maneuvers

- Rectangular course:
 - Note: this can be assessed while doing traffic patterns.
 - Student says, student does.
 - Verify correct procedure.
 - Verify proper wind correction on the various pattern legs.
 - Standards:
 - Enter a left or right pattern, 600 to 1,000 feet above ground level (AGL) at an appropriate distance from the selected reference area, 45° to the downwind leg.
 - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
 - Maintain altitude ±100 feet; maintain airspeed ±10 knots.
- S-turns:
 - Student says, student does.
 - Verify correct procedure.
 - Both turns must have the same radius.
 - Bank must be adjusted for ground speed/winds.
 - Standards:

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- Enter perpendicular to the selected reference line, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
- Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
- Maintain altitude ± 100 feet; maintain airspeed ± 10 knots.
- Turns around a point:
 - Student says, student does.
 - Verify correct procedure.
 - Radius must be the same throughout the turn.
 - Bank must be adjusted for ground speed/winds.
 - Standards:
 - Enter at an appropriate distance from the reference point, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
 - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
 - Maintain altitude ± 100 feet; maintain airspeed ± 10 knots.

Navigation

- Pilotage and dead reckoning
- Navigation systems and radar services
- Diversion
- Lost procedures

Slow Flight and Stalls

- Maneuvering during slow flight:
 - Student says, student does.
 - Verify correct procedure.
 - Ensure the student is pitching for airspeed and using power to control altitude.
 - Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Establish and maintain an airspeed at which any further increase in angle of attack, increase in load factor, or reduction in power, would result in a stall warning (e.g., airplane buffet, stall horn, etc.).
 - Accomplish coordinated straight-and-level flight, turns, climbs, and descents with the airplane configured as specified by the evaluator without a stall warning (e.g., airplane buffet, stall horn, etc.).
 - Maintain the specified altitude, ± 100 feet; specified heading, $\pm 10^\circ$; airspeed, $+10/-0$ knots; and specified angle of bank, $\pm 10^\circ$.

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- Power-off stalls:
 - Student says, student does.
 - Verify correct procedure.
 - Ensure proper coordination.
 - Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Configure the airplane in the approach or landing configuration, as specified by the evaluator, and maintain coordinated flight throughout the maneuver.
 - Establish a stabilized descent.
 - Transition smoothly from the approach or landing attitude to a pitch attitude that will induce a stall.
 - Maintain a specified heading $\pm 10^\circ$ if in straight flight; maintain a specified angle of bank not to exceed 20° , $\pm 10^\circ$ if in turning flight, while inducing the stall.
- Power-on stalls:
 - This can either be done in the takeoff or climb configuration.
 - Student says, student does.
 - Verify correct procedure.
 - Ensure proper coordination.
 - Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Establish the takeoff, departure, or cruise configuration, as specified by the evaluator, and maintain coordinated flight throughout the maneuver.
 - Set power (as assigned by the evaluator) to no less than 65 percent power.
 - Transition smoothly from the takeoff or departure attitude to the pitch attitude that will induce a stall.
 - Maintain a specified heading $\pm 10^\circ$ if in straight flight; maintain a specified angle of bank not to exceed 20° , $\pm 10^\circ$ if in turning flight, while inducing the stall.
- Spin awareness:
 - Coordination!

Basic Instrument Maneuvers

- The student must be able to perform the following:
 - Straight and level flight.
 - Standard rate turns and medium level turns.
 - Constant airspeed climbs and descents.
 - Recoveries from unusual attitudes (ensure that the student recovers using the correct procedure depending on nose low or nose high).
- Standards:
 - Maintain altitude ± 200 feet, heading $\pm 20^\circ$, and airspeed ± 10 knots.

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- Level off at the assigned altitude and maintain altitude ± 200 feet, heading $\pm 20^\circ$, and airspeed ± 10 knots.
- Turn to headings, maintain altitude ± 200 feet, maintain a standard rate turn, roll out on the assigned heading $\pm 10^\circ$, and maintain airspeed ± 10 knots.
- Recognize unusual flight attitudes; perform the correct, coordinated, and smooth flight control application to resolve unusual pitch and bank attitudes while staying within the airplane's limitations and flight parameters.

Emergency Operations

- Emergency descent
 - Student says, student does.
 - Verify correct procedure.
 - Standards:
 - Maintain orientation, divide attention appropriately, and plan and execute a smooth recovery.
 - Use a bank angle between 30° and 45° to maintain positive load factors during the descent.
 - Maintain appropriate airspeed $+0/-10$ knots, and level off at a specified altitude ± 100 feet.
- Emergency approach and landing
 - Student says, student does.
 - Verify correct procedure.
 - Evaluate choice of landing point.
 - Landing must be made with a head wind.
 - Ensure that the student verifies with the checklists.
 - Standards:
 - Establish and maintain the recommended best glide airspeed, ± 10 knots.
 - Select a suitable landing area considering altitude, wind, terrain, obstructions, and available glide distance.
 - Plan and follow a flightpath to the selected landing area considering altitude, wind, terrain, and obstructions.
 - Complete the appropriate checklist.
- Systems and equipment malfunctions
 - Give the student a scenario such as:
 - Low oil pressure, high oil temperature, carburetor icing, lost comms, etc.
 - Ensure that the student can troubleshoot and use the emergency checklist.
- Emergency equipment and survival gear
 - This should be briefed/quizzed on the ground.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Homework:
 - Study for the end-of-course evaluation.
 - Plan one of the required TOLD card and cross-country problems.

Completion Standards

The student shall demonstrate proficiency that meets or exceeds the standard of performance outlined in the current FAA Private Pilot Airman Certification Standards. Any deficiency will require further instruction until those standards are met.